Residual Stress Reduction via Orientation Changes of Powder Bed Fusion Parts

A Project Presented to The Faculty of the Department of Mechanical Engineering
San Jose State University

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Abstract

Additive Manufacturing (AM) is a relatively new method of manufacturing in which material is added layer by layer. Initially this happened for resins and plastics, with Vat Photopolymerization (VPP) being one of the first methods. This method involves resins fusing as a result of being exposed to light or lasers. Fused Deposition Modeling (FDM) stacks extruded plastics on top of each other to create parts. More recently, metals have been added to the library of materials that can be made into parts via AM.

Powder bed fusion is a method of three-dimensional (3D) printing metals that has risen to prominence recently since its cost has come down. Companies like Czinger Automotive have combined topology optimization and metal 3D printing to create performance parts that are stronger and lighter than what is possible through traditional manufacturing methods. It has also found its way into aerospace; making turbine parts out of superalloys to be sent into the harshest environments imaginable.

The most common form of metal 3D printing is Laser Powder Bed Fusion (LPBF). This is a process in which powdered metal is fused together layer by layer with a laser to produce a monolithic part. However, the fusion process leaves behind residual stresses in the finished part due to the uneven distribution of heat in the powder and melt pool.

The goal of this project is to create a tool that can minimize the residual stress in a part by finding the ideal orientation in 3D space to print the part in question. To accomplish this, Solidworks will be used to rotate parts, and ANSYS will be used to find the residual stress in the part in its particular orientation. To control these two programs, the tool will be created in Python. This tool will be able to rotate the part in Solidworks, find the residual stress in ANSYS, and intelligently iterate to find the orientation with the least residual stress.

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1 Introduction

1.1 Literature Review

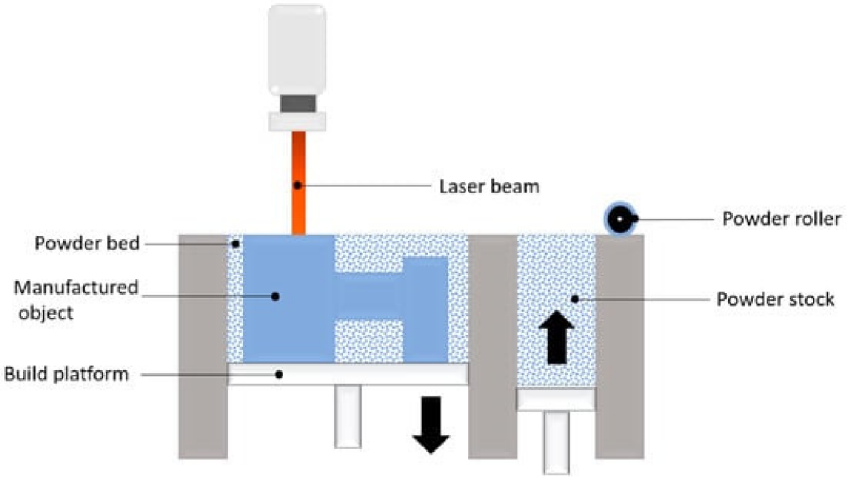


Figure 1: Schematic of LPBF, from [1]

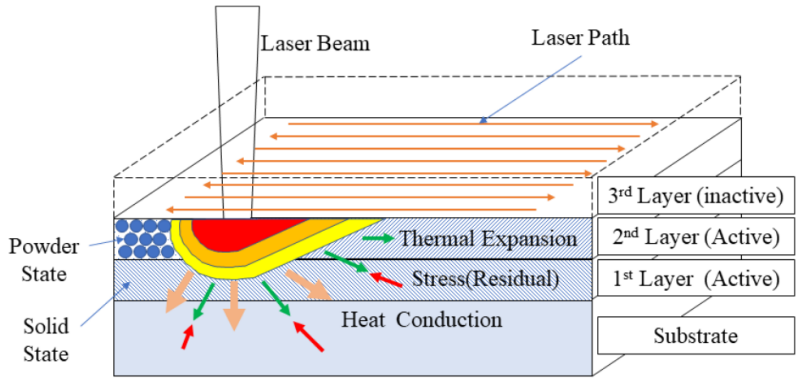


Figure 2: Schematic heat and its effects on stress during LPBF, from [2]

When printing a part, through powder bed fusion (see Figure 1) the laser traces a path around the bed of powder before new powder is added on top to start the next layer. However, when heating the powder, the laser does not heat evenly; the center of the beam is narrow, but a wider area than just the beam is heated, but to a lower temperature than the center of the beam [1], [3]. This difference in temperature means that different parts of the part will solidify at different times, and the differences in thermal expansion through this delta in effect stretches the part before it's even complete. This causes residual stress (see Figure 2). The cold parts of the metal constrain the warm parts from cooling to a natural shape, pulling it in unwanted directions [4]. The three parts of this literature review are: Residual Stresses in Powder Bed Fusion, How Print Orientation Affects Stress, and Simulation of Metal AM.

This was decided to cover the topic of residual stress, the factors that affect it, and how to simulate these stresses.

1.2 Residual Stresses in Powder Bed Fusion

Investigations into the stresses present in metal 3d printed parts have yielded valuable information. Fojtik et al. [5], investigated the stresses present in a 3d printed brake pedal. The most visible test of the stress was when they cut a part of the pedal. The split piece opened up outwards like a flower, indicating that was the position it desired to be in; the state of lowest stress. They also concluded that support materials produce additional residual stress in the areas immediately around them, due to the constant heat cycling that building the support creates, and the support's role in locking the part into shape, rather than letting it expand and contract naturally.

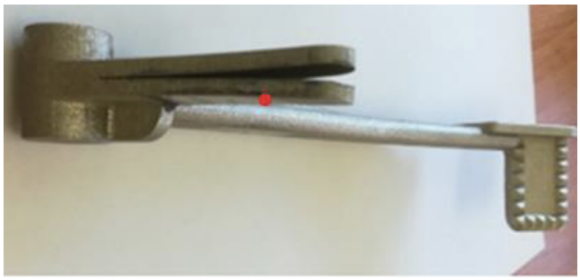


Figure 3: Deformation of a pedal printed by LPBF as a result of residual stresses, from [5]

An interesting finding from Park et al., was that the residual stresses on the last surface to be printed are tensile. The mechanic behind this is; the lower layers cool and shrink, but the top layer gets stuck on the layer below. The molten metal bonds to the solid layer below, but when it contracts while solidifying, it is unable to contract as it naturally would [6]. However, once heat treated, the surface changed to compressive stress [6]. Depending on the application, this can be highly desirable, akin to shot peening in use.

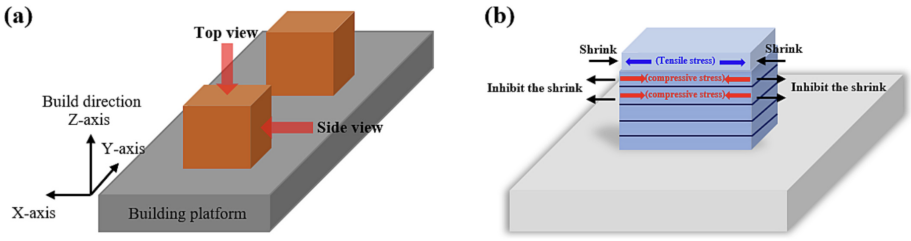


Figure 4: Illustration of the stress-causing shrinkage, from [6]

1.3 How Print Orientation Affects Stress

One of the methods that is under study to remedy the problems of residual stress in a 3D print is to change the orientation of the prints. Yang et al. [7], studied the effects of

differing print orientations for parts printed via powder bed fusion. This group focused on the anisotropic effects of printing in layers with metal. Their experiments yielded results akin to the anisotropy in a traditional FDM part. The print layers are not as well bonded together as opposed to the metal made in one layer. On the topic of building layers, Fojtik et al., studied the hatch pattern between layers and found that changing the raster pattern by 45 degrees between print layers reduces residual stress [2]. Takazawa et al. [8], studied the methods to reduce warpage in a printed part by manipulating the orientation of the hatches and the infill density. By optimizing these two parameters, part warpage is decreased, inferring that there is a reduction in residual stress as well; as the part does not bend on its own [8].

1.4 Simulation of Metal AM

Simulation of parts made via 3D printing can be accomplished using softwares like ANSYS. Alzyod and Ficzere [9], [10] experimented with the simulation techniques available in traditional 3d printers in two papers (FDM, SLA, etc.). They found that the residual stresses can be accurately predicted using simulation combined with empirical data. In their study, they ran simulations, printed parts, tested them, and went back and tuned the simulation based on the stresses in the real parts. This allows for accurate simulation of more complex parts based on the simulations that have been tuned. Zelder et al. [11] trained a PointNet architecture to optimize part orientation for printing by LPBF. This work is optimized for cost, geometric accuracy, and print time. Residual stress was mentioned, but not explored as one of the variables in the process. Their parts were rotated in space, attempting to minimize support structures, z-height (which increases time) and keep geometric stability (like errors based on layers in the AM process) [11].

1.4.1 ANSYS Analysis

Weinhold et al. [12] ran a manual version of this optimization problem for their conference paper. To take advantage of the 3D printing process, they used topology optimization to find the minimum amount of material needed to build a functional connecting rod for a 212cc engine. Then, they ran simulations at three orientations: 0, 45, and 90 degrees to find which had the least distortion. Their paper found that 0 degrees of rotation (horizontal) had the least distortion (and by extension residual stress) [12].

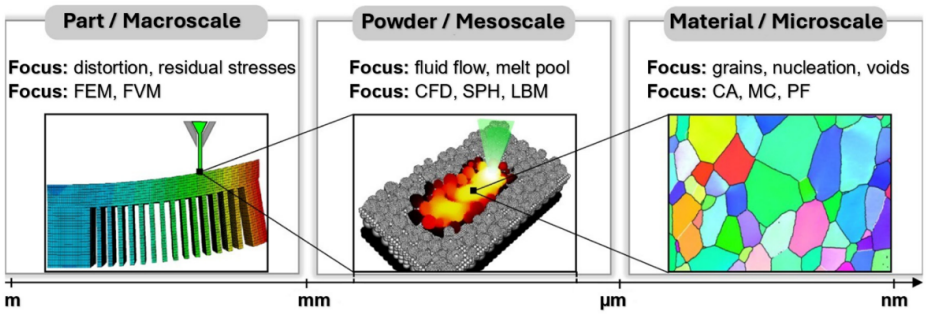


Figure 5: The three scales of LPBF simulation, from [13]

Kaskac et al. [13] describe the different types of LPBF analysis in their paper, as shown in Figure 5. Microscale focuses on heating, voids, etc. at the grain structure level. Mesoscale looks at the scale of the melt pool (0.01-1.0mm) and macroscale looks at the part level. This report focuses on the macroscale, as does ANSYS.

Kaskac et al. [13] makes some notes about the ANSYS method of simulation in their paper. ANSYS simulates each individual voxel layer in the mesh as one scan layer, and there are three levels of simulation for LPBF: assumed uniform stress, scan pattern, and thermal stress (thermal stress will be used for this paper). They also note that ANSYS does not always take into account the moving heat source when the dominant thermal gradient is in the opposite direction [13].

1.4.2 Analysis of Other AM Methods

Another method of manufacturing metal parts is through direct energy deposition. Rong et al. [14] have done work in predicting the residual stresses associated with this method of manufacture through FEA. To accurately model the material that is produced from melting through cooling, the different phases through which a material (steel) will pass through are added to the FEA model [14]. While cooling, phases such as martensite and bainite are formed, and they each have different rates of cooling, thermal expansion, etc. So incorporating the different possible phases into the model will increase its accuracy over just having a single phase being cooled [14]. The process of direct energy deposition is different from LPBF, however, the principles that create the residual stress are the same.

1.5 Status and Outlook

The field of residual stresses in powder bed fusion prints is one undergoing a plentiful research. It seems that the underlying causes are reasonably well understood. Lab work specifically in reducing the residual stress seems somewhat lacking, however. As shown in the paper by Park et al. [6], having broad, flat surfaces normal to the direction of printing bed travel causes high residual stress buildups. One avenue to explore would be diagonalizing the print to reduce this stress. At a minimum, this would reduce the tensile stress lengthwise along the top print surface, however, this would cause new stress patterns to show up in the cross section of the part.

Work has been done using deep learning to optimize part print orientation, but without concerning residual stresses. These different research instances can be used together to optimize for the residual stress and print orientation problem. The pattern of laser travel has also been studied, but it doesn't seem like part orientation in space has garnered much attention yet. An investigation into the orientation of prints within 3D space will complement the work that has already been done into surface stresses and raster orientation in prints.

2 Methodology

2.1 Geometry Manipulation

To simulate different geometry orientations, appropriate CAD must be produced with the rotated part. To do this, Solidworks [15] was used. This is the simplest way to generate new CAD from an existing model.

Solidworks has an Application Programming Interface (API) [16] that allows Visual Basic for Applications (VBA) to be passed from a Python script running on a computer into Solidworks. Using this, a feature can be added to the existing feature tree of any part that is in the native Solidworks part format. In the case of this project, the "Rotate" command can be added in to rotate a named part.

To export the part from Solidworks to a format that ANSYS can use, such as .step, another VBA command is used. This one saves the part as a Step file. A string can also be passed into the VBA script to change the naming of this file, which is useful for organization and file handling between scripts.

2.2 Controlling a Simulation in ANSYS

To automatically simulate LPBF in ANSYS, the PYMECHANANICAL and PYADDITIVE [17] [18] Python libraries can be used. These libraries function as an API between Python and ANSYS. Python needs to find or be shown the file path that contains the file that opens ANSYS Mechanical. Once it has this, Python can open ANSYS and send commands into the opened instance of ANSYS.

2.3 LPBF Simulation in ANSYS

There is a caveat that LPBF residual stress simulation is not available in PYANSYS. The workaround for this issue is to use a template file. This is created by running an LPBF simulation, then saving it as a template file. The original simulation is inconsequential and will be overwritten, but it is important that the original simulation is correct and complete.

2.4 Algorithms

To implement the algorithms mentioned in the research section, a higher level Python script must be used. This high-level script controls the rotation and simulation scripts. The first action this script takes is to run initial simulations. The goal of these is to establish a baseline stress vs. orientation relationship for the searching algorithm to build off of. For this simulation, a Sobol sequence is used. As explained in Renardy et al., the Sobol sequence generates a quasi-random set of numbers. [19]

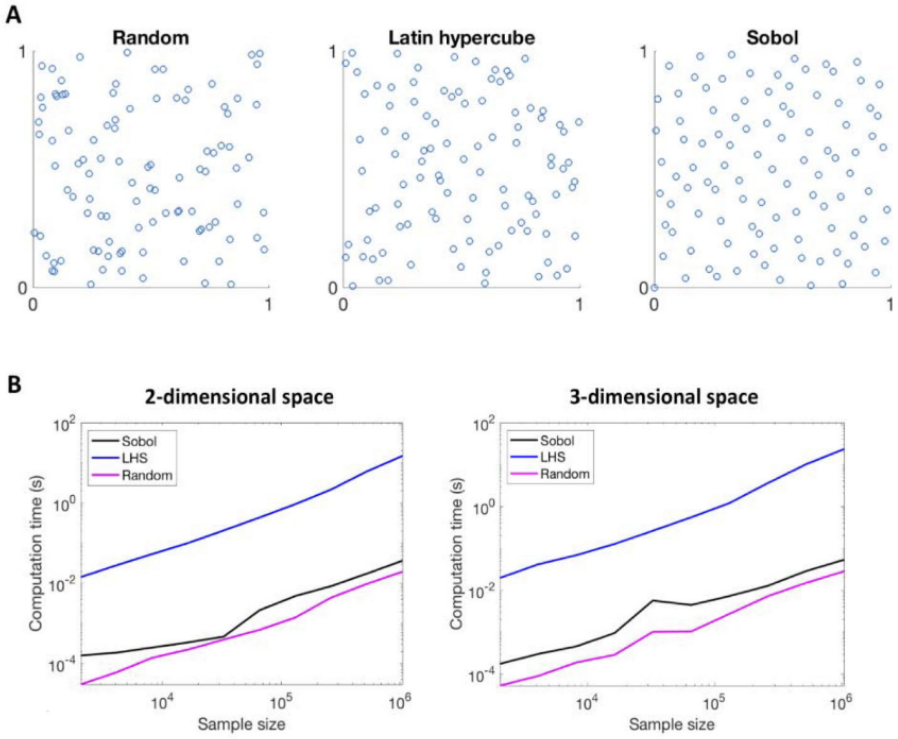


Figure 6: Comparison of Sobol Sequence to other number generation methods [19]

In this instance, it is implemented via a Python library. The Sobol Sequence gives a set of numbers (x,y,z) between 0 and 1. Then the number is multiplied by 180 degrees to give an amount of rotation. This is done for the first few "seed" simulations, which are designed to get the baseline for the simulation going. The Sobol algorithm is used because of its capability for good coverage of the randomized space. Although this simulation takes place in the 3D space, ANSYS does its simulation layer by layer. Meaning that everything along one Z-coordinate is taken as one step; raster angle does not matter. Because of this, rotation about the Z-axis is removed, saving iterations and compute time. [20]

Once the initial stresses are known, the high-level script runs an algorithm to come up with new angles to simulate. These angles are passed to the rotation script, and the rotated file is then simulated. This cycle happens until the criteria for low residual stress is met. Then the final angle is recorded, and the step file is to be printed.

Gradient descent takes too many iterations, full contour optimization also takes too many iterations; it is Gaussian A related study on optimization of prints was run by Menon et al.. In this paper, the computational expense of running simulations was, leading to the use of a Gaussian Process. [21]

2.5 ANSYS LPBF Residual Stress Simulation

According to the ANSYS LPBF Simulation Guide [22], "The simulation is not meant to provide detailed thermal or structural results needed for prediction of micro-level process phenomena". Thus, it is not meant for detailed looks at the crystal-level happenings of a

part printed via LPBF.

The LPBF simulation occurs following the same pattern as the printing of a part. It starts at the build plate and builds the model up, layer by layer using "Standard element birth and death" [22]. ANSYS doesn't simulate the scan path of the laser, as it "Would take enormous compute time making it impractical" [22]. Instead, it applies heat to entire layers. This saves compute time, while still giving an indication of the residual stress generated.

ANSYS lumps layers into "Super Layers" [22], clumping layers together assuming that the layers are similar enough in profile that the heating effects will be similar between them. These layers are heated as they are generated, and appropriate heat is retained in layers below to give the thermal gradients responsible for stresses.

Table 1: Simulation parameters used in ANSYS LPBF simulations [22].

Parameter	Meaning
Base Geometry	Sets Print Base
Material	Material
Mesh Criteria - Mesh Method	Voxel, Tetrahedral, Cartesian
Supports	Sets up supports for the part
Heating Method	How is heat applied to a layer (Melting Temperature or Power)
Scan Time	Amount of time in layer
Beam Power	Power of Laser
Absorptivity	Fraction of laser power that goes into heating
Hatch Spacing	Space between scans
Deposition Thickness	Depth of powder per recoat
Scan Speed	Speed of scan
Dwell Time	Duration of pause between layers
Preheat Temperature	Starting temperature of build plate
Base Thermal Boundary Condition	Where the build plate is heated
Base Structural Boundary Conditions	Where base is fixed
Gas/Powder Temperature	Temperature of chamber and powder
Gas Convection Coefficient	Convection coefficient between new top layer and chamber gas
Powder Convection Coefficient	Effective convection coefficient from sides of build to powder bed
Powder Property Factor	Knockdown factor to estimate powder properties

The parameters and their descriptions in Table 1 are the values that are required to run this simulation as described in [22]. These are the parameters that are to be set before running the LPBF simulation. These values vary between alloy and printer, so it is important to get them right, or the simulation will be inaccurate to the print.

$$[H]\begin{bmatrix}[M] & [0]\\ [0] & [0]\end{bmatrix}\begin{Bmatrix}\ddot{u}\\ \ddot{T}\end{Bmatrix}+\begin{bmatrix}[C] & [0]\\ [C_{tu}] & [C_t]\end{bmatrix}\begin{Bmatrix}\dot{u}\\ \dot{T}\end{Bmatrix}+\begin{bmatrix}[K] & [K_{ut}]\\ [0] & [K_t]\end{bmatrix}\begin{Bmatrix}u\\ T\end{Bmatrix}=\begin{Bmatrix}F+[K_{ut}]\{T_{ref}\}\\ Q\end{Bmatrix} \quad (1)$$

Jung et al. [2] use Equation 1 to describe the background of what is happening in LPBF residual stress FEA. It couples heating and cooling of the powder in the build to the expansion and contraction of the part being printed. $[K^{ut}]$ is the thermoelastic stiffness matrix [2]. $[C_{tu}]$ is the attenuation matrix, and $[K^t]$ is the heat conduction matrix. $[K^t]$ is found by adding ($[K^{tb}]$) (Conductivity to material diffusion) and ($[K^{tc}]$) (conduction based on the material convection surface).

2.6 Geometry Modifications Necessary for Use

There are several geometric and logistical requirements needed to prepare the geometry for simulation with this software. For the AM simulation within ANSYS, both the part being printed and the print bed need to be present in the imported geometry. In this simulation, it is required that they are named "base" and "part" with base being a flat rectangle, and part being what is being printed. This is necessary for Solidworks to identify which part to move, and for ANSYS to find which part to simulate. The rectangular base must have its bottom face (away from part) located at z=0. This is so a fixed constraint can be applied to constrain movement degrees of freedom during the mechanical simulation. The part must be offset in the z from the build plate to allow for supports to be built, otherwise the part will be stuck to the build plate, and its geometry will be affected during the wire EDM separation of part from build plate.

3 Semester 1 Work

3.1 Explanation of Scripts

3.1.1 Solidworks Manipulation Script

Main file
The main code of this section of the tool. It calls other files that are written in Python, packages them as VBA scripts, and then sends them to Solidworks to execute.
SWOPEN

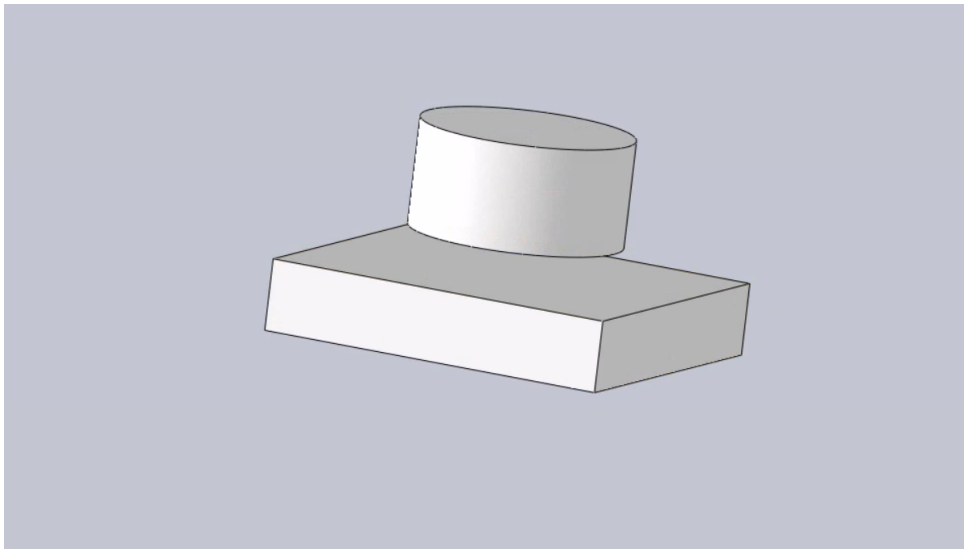


Figure 7: The Solidworks screen when the initial file is opened.

This code finds the instance of Solidworks installed on the local machine, and opens it. Then, it opens a step file. The path of this step file is currently named in the SWOPEN file, but will be moved to be a variable passed from the main file, so it is easily editable by a user. Figure 8 shows Solidworks just after SWOPEN is run; an unmodified part geometry open.
SWROT
This code rotates the part in question. Figure 8 shows Solidworks as it is after SWOPEN runs. It currently requires the solid body being rotated to be named "part". This code finds the running instance of solidworks and attaches to it. Then, it finds the solid body named "part". It inserts a move feature into the feature tree and assigns it to "part" and rotates the part by a set angle. This angle is passed into the script from main when it is called. Right now, the center of rotation is the origin of the Solidworks document. This will be changed later on. Figure 9 shows the Feature Tree after the rotation feature has been inserted.

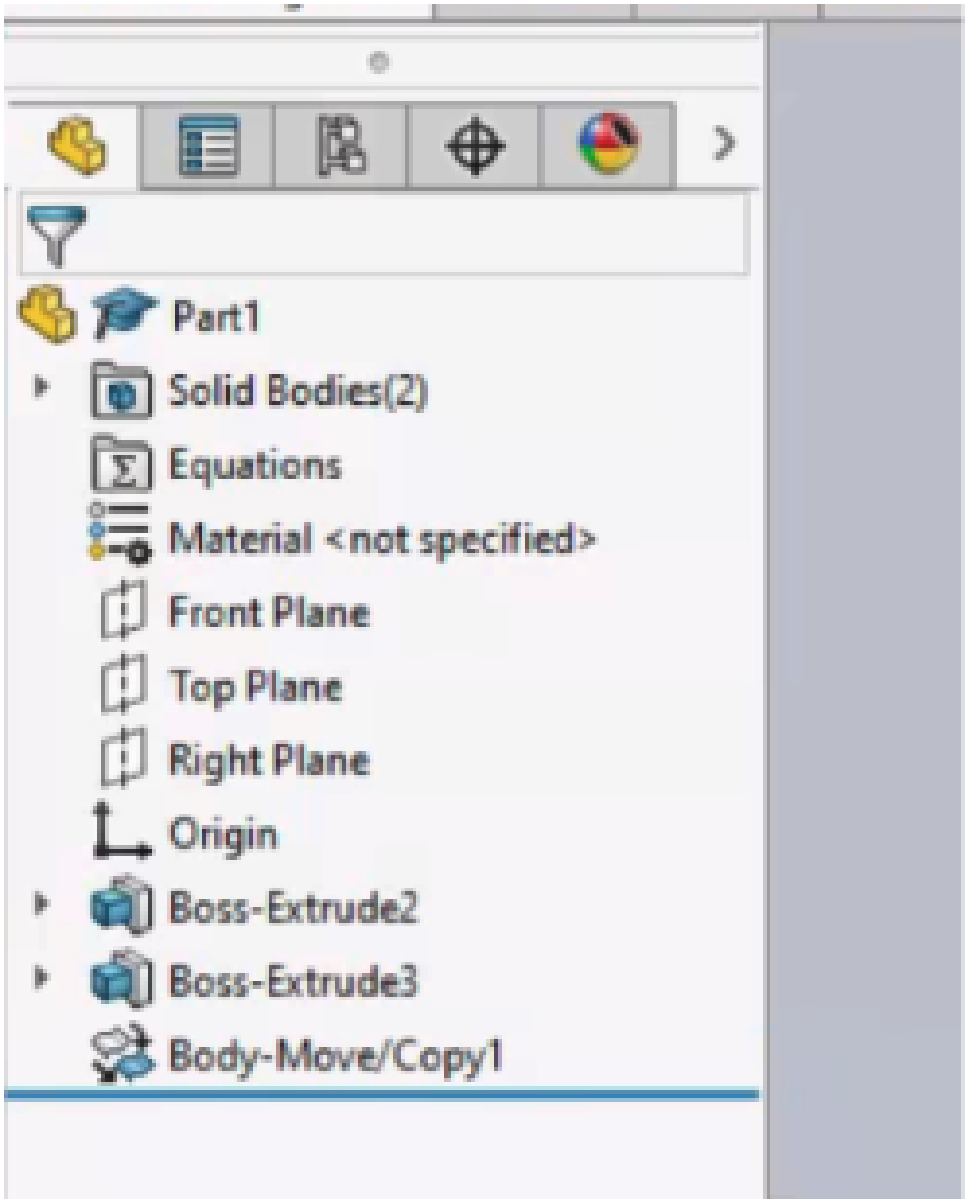


Figure 8: The feature tree in Solidworks, showing the inserted feature to rotate the part.

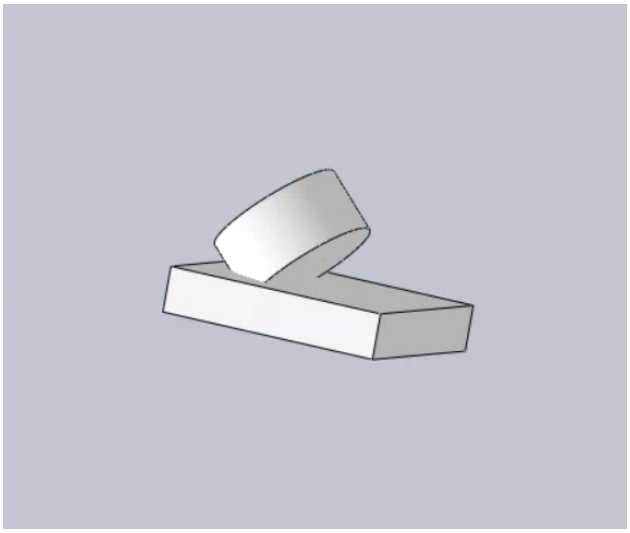


Figure 9: The part as rotated by SWROT. Note the interference.

ANSYS requires the build plate to be located at Z=0, so rotating a part about the origin of the document is not a good idea. It will cause the part to collide with the build plate. Future updates to the code will have logic to find the center of gravity of the part being rotated, and rotate it about that, or interference detection along with moving the part until it doesn't interfere with the build plate. Figure 10 shows the part that has been rotated, and displays the collision that typically occurs.

SWEXPORT

This script exports the modified document out of Solidworks as a step file. It takes in the naming convention as an input from main and appends it to the file name when saving the file.

3.1.2 ANSYS Execution Script

Main

The main is responsible for opening the instance of ANSYS mechanical, and calling the following scripts to run the simulation. **Call Template** This script opens the template for LPBF that is included in the folder for the code. The template needs to exist because the coupled simulation that ANSYS uses for residual stress in LPBF is not available to call from the PYANSYS library. To get around this limitation and keep the code all in Python, the template was used, and edited with the Python script. This does lead to a bit of artifacting, but the results are not affected. Delete Geom Because a template was used for this Python script, there is a bit of cleanup that needs to be done. The template needed to have a geometry and finished simulation inside, so that needs to be removed. The Delete Geom script deletes the preexisting geometry of the original simulation in the template to make room for the new one. Import Geom This script imports geometry from a step file. Right now the file is hard coded into the import geom document, but an upcoming improvement is passing the step file from main, which receives it from the Solidworks part manipulation script. It also assigns the imported bodies to "part" and "base" within the AM simulation

process. This script then deletes the existing mesh controls from the template simulation.
Mesh

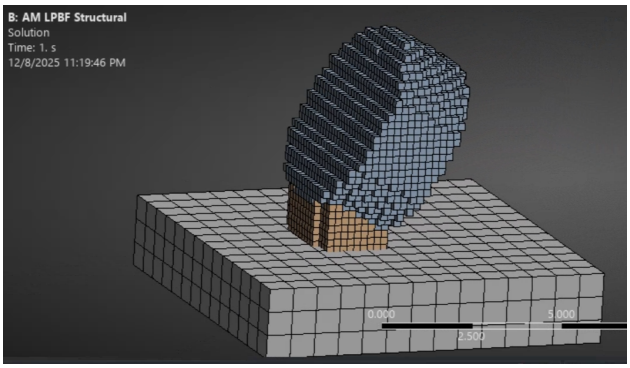


Figure 10: The meshed part; note the voxelized mesh.

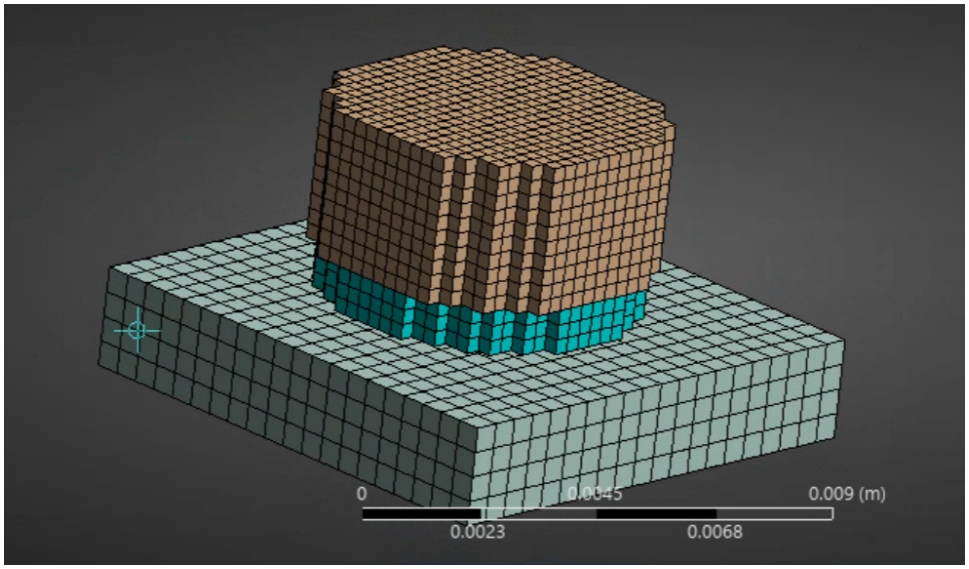


Figure 11: The auto-generated supports.

This script generates commands ANSYS to generate a mesh for the imported geometry. It starts off by defining the global mesh size, then assigns refinements and methods to certain geometries based on the part names. Then, ANSYS meshes the parts; the meshed setup is shown in Figure 11. Once this is done, the script deletes the supports that were present for the deleted geometry and defines the parameters for the supports that the new geometry will use. Figure 12 shows the supports as generated by the script. Temp and Run

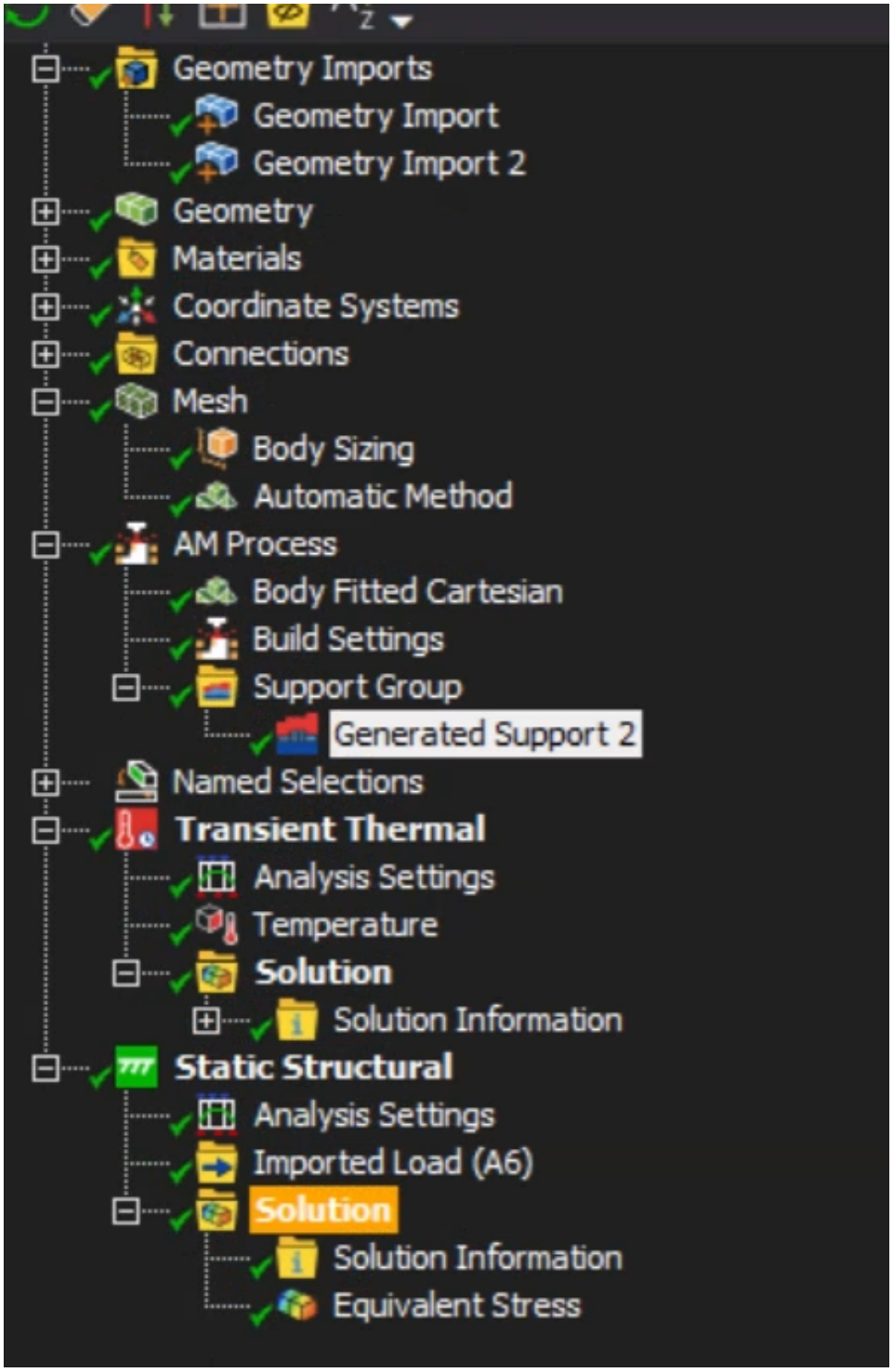


Figure 12: The feature tree as modified by the script.

Temp and run is the final script in the running of the simulation. It sets the temperature of the thermal aspect of the simulation, then commands the simulation to solve. Figure 13 shows the feature tree of the script, including the Temperature and Static Structural that

make this a coupled simulation. Under Geometry Imports, it can be seen that there are two items; this is a result of using the template; the geometry was deleted from the first import, and the geometry from the second was used.

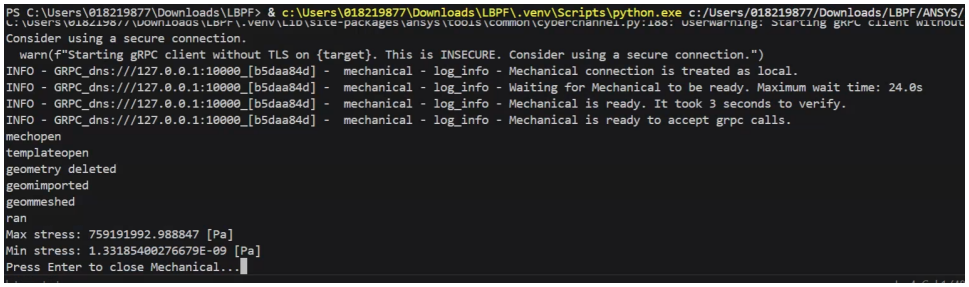


Figure 13: The terminal after the script has run.

Figure 14 shows the terminal as it appears while the ANSYS script is running. There are checkpoints at which certain strings are printed to let the user know that the script is working correctly. Currently, the main script receives the stress values from the simulation and shows them.

4 Results

4.1 Puck

This part was chosen as a first simulation because of its simplicity. The puck acts just as a part that can be simulated to find out whether the simulation setup and the refinement algorithm work. There is no industrial motivation for simulating this part, just a need to verify that all scripts work and that the simulation will run without crashing.

4.1.1 Setup

The puck was constructed with a diameter of 5mm and a height of 3mm. The voxelized mesh was created to balance run time and fidelity to the original shape. Thus, a mesh of 0.5mm was used for the part under test, and the "base" was given a coarser mesh of 2mm.

4.1.2 Results

An image of the final stress result is shown below.

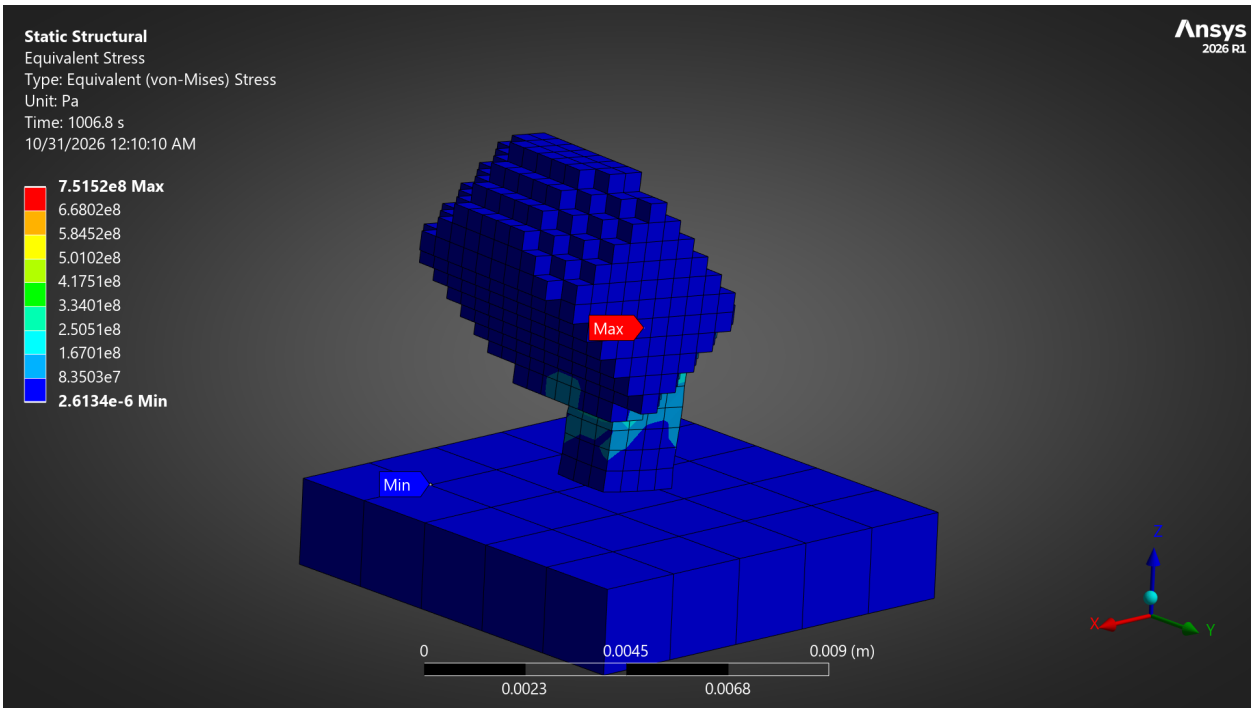


Figure 14: The final stress result of the simulated puck.

This was the final rotation after a full run of stress optimization.

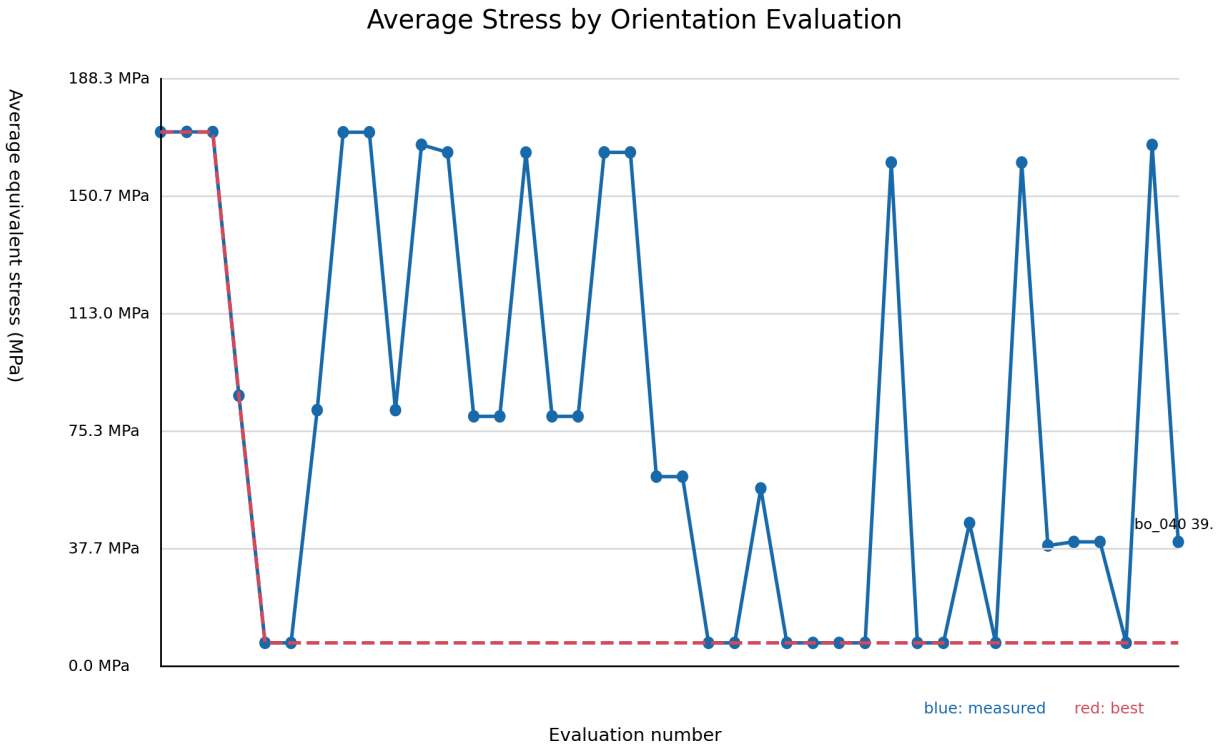


Figure 15: The graph showing the progression of average stress over the run's iterations.

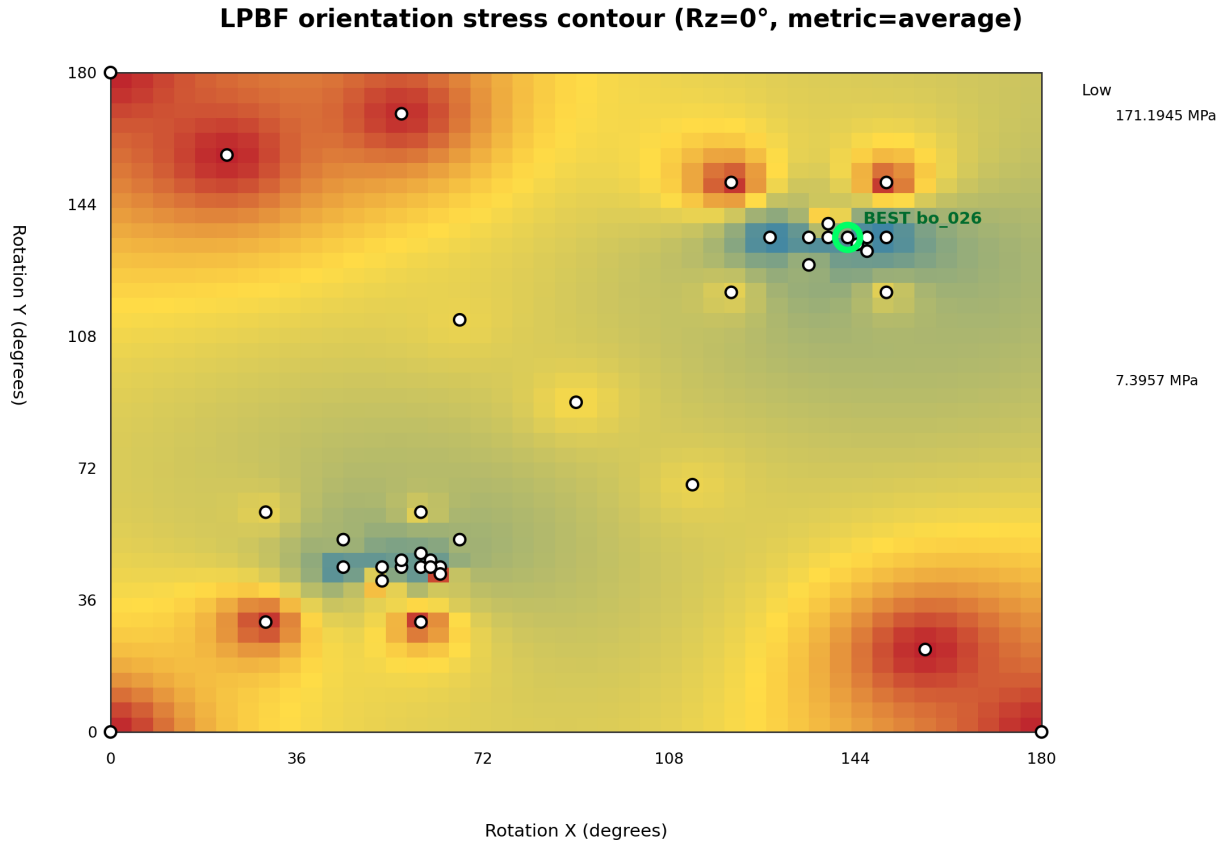


Figure 16: A contour map of the stress vs. rotation angle. Note the data is quite sparse, so graph usefulness may be questionable.

For this run, when looking at 15, it can be seen that the stress bottomed out relatively early in the run. The algorithm took this location into account, and continued to run the original "seed" simulations, to get an awareness of the design space. Once those runs were done, the search went back to the area around the minimum stress already found, resulting in 14. Then it searched around the area. The step is 15° and the program explored around that area. It found a few rotation points that were higher, and moved away from them. Eventually, after a number of iterations (find that info) around the minimum stress point, there was no notable improvement. Thus, the criteria was met to end the simulation (find this out too).

Through this simulation, it was found that the optimal orientation is a rotation of $\theta_x = 61°$ and $theta_x = 45°$ to have the optimal stress. This result could have been found quicker than this graph shows if the criteria were loosened, but that is not always the case. There is also a sort of contour map in

4.2 Compressor Wheel



Figure 17: A 3D printed compressor wheel [23].

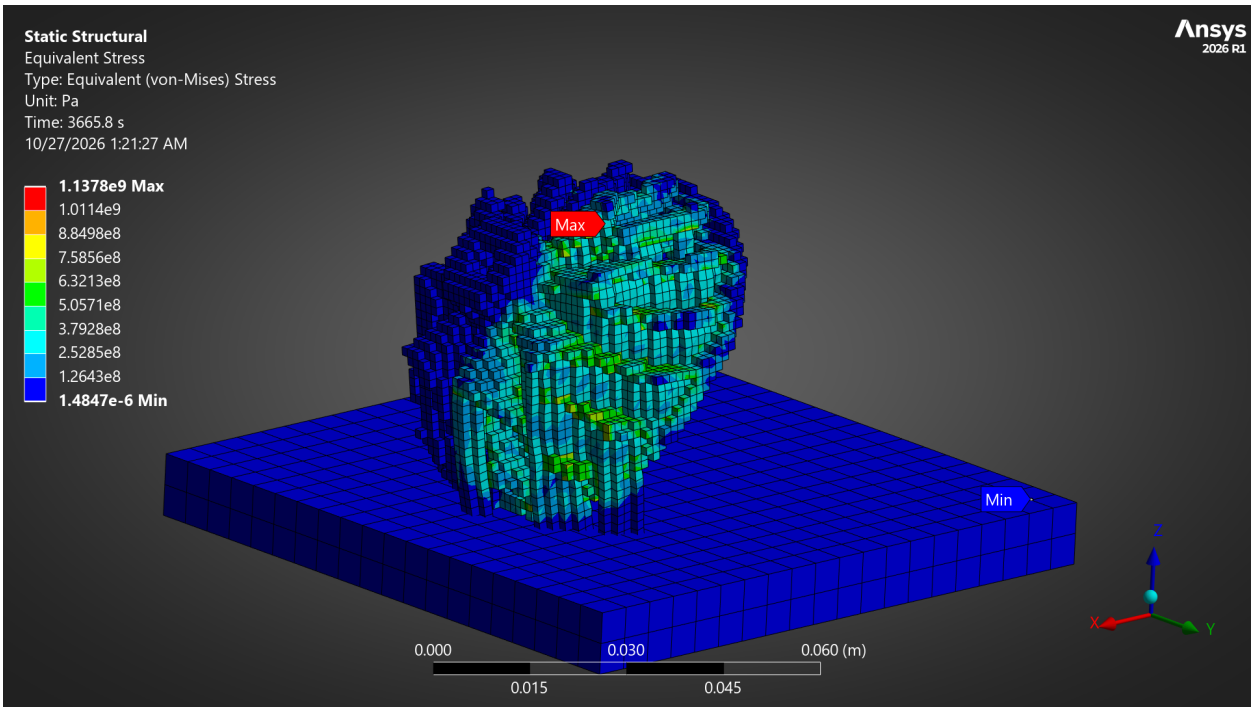


Figure 18: The final stress result of the simulated compressor wheel.

This was the final rotation after a full run of stress optimization.

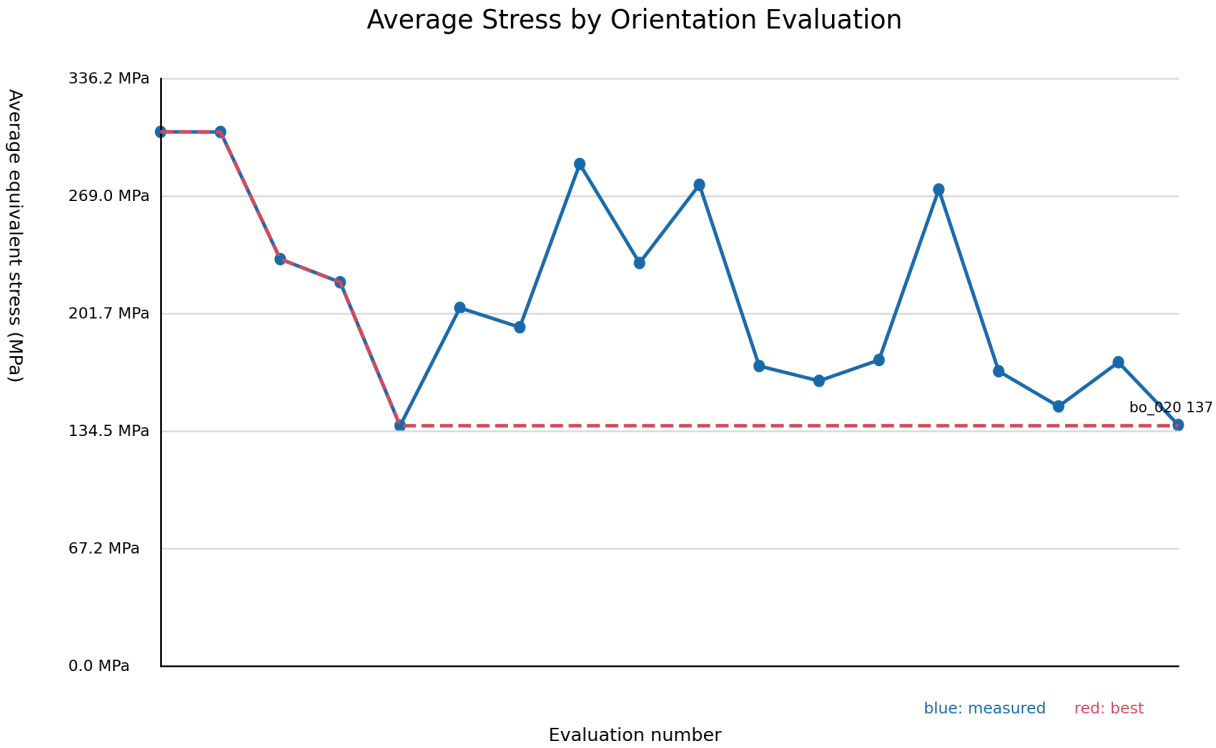


Figure 19: The graph showing the progression of average stress over the run's iterations.

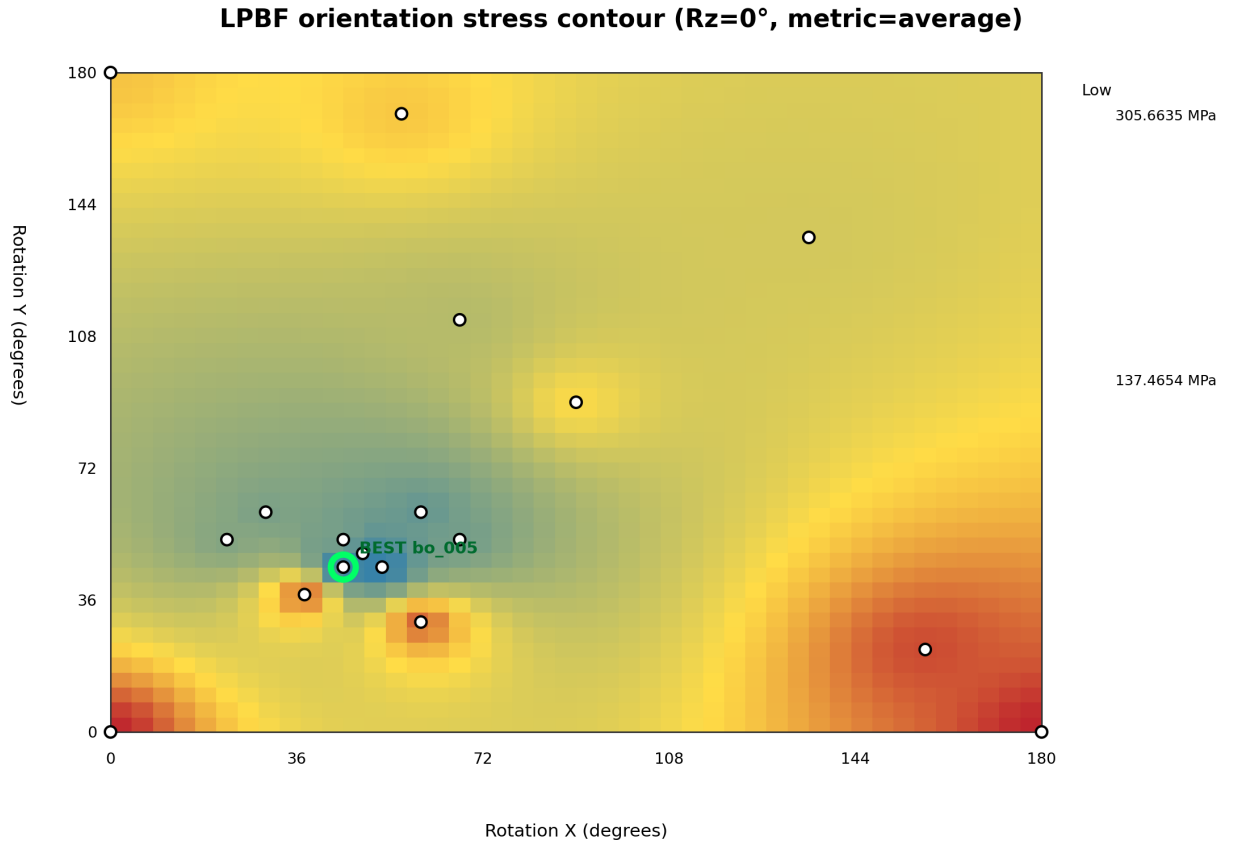


Figure 20: A contour map of the stress vs. rotation angle. Note the data is quite sparse, so graph usefulness may be questionable.

4.2.1 Motivation

Compressor wheels are an ever present part of air movement. They most frequently show up in turbochargers for automotive purposes, but also make appearances in small jet engines. They are frequently cast or machined. Both of these processes are not optimal for mass manufacturing. Machining these parts requires a large piece of round billet to start, out of which the majority of material is machined away, taking up significant machine time and wasting material. Casting this part requires investment casting, due to the amount of overhangs in the part. Investment casting has a large capital overhead and a high amount of tooling investment. The complex geometry and demands of spinning at 100,000 rpm brings AM into the question for this part. The geometry simulated in this section was found on Grabcad, and is a stand-in for production geometry to show the capabilities of this software.

4.2.2 Setup

To set up this run, the geometry steps explained in Methodology were used. Once the geometry was prepared, the run was started with a mesh of 1.5mm on the compressor wheel, and 5mm on the base.

4.3 Turbine Blade

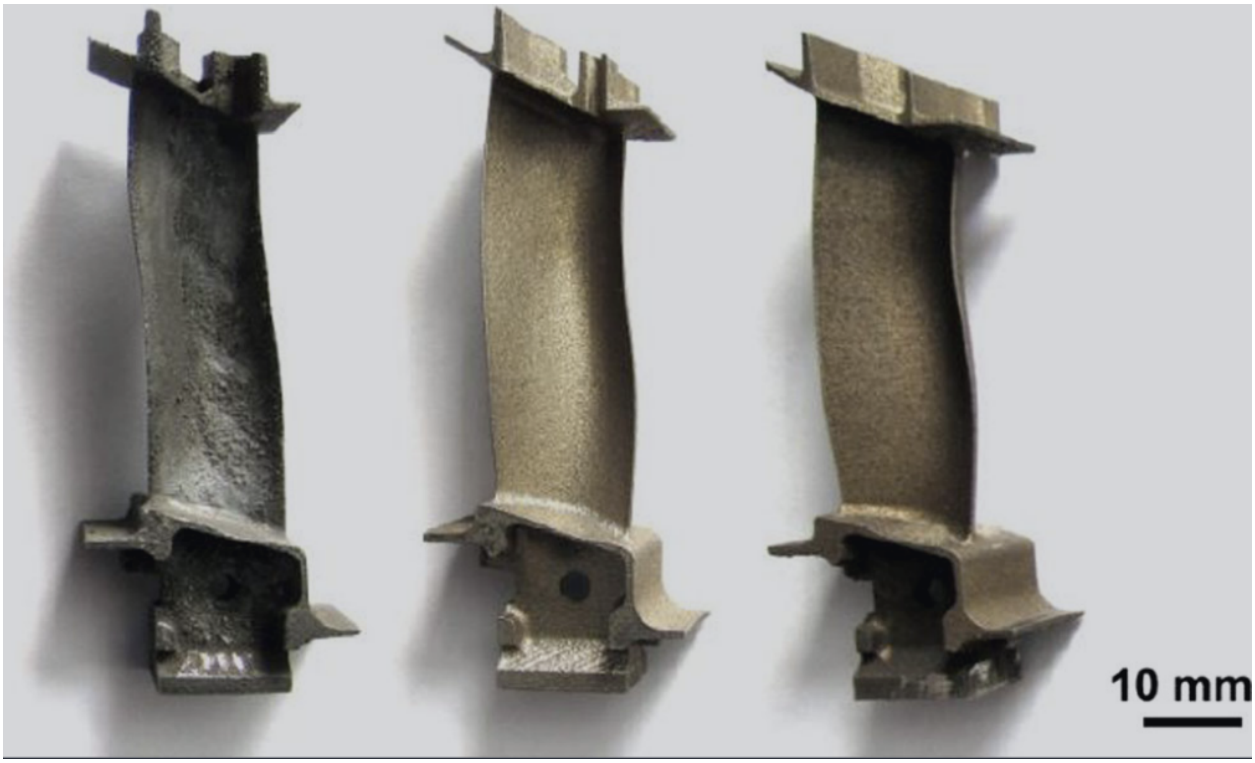


Figure 21: A 3D printed turbine blade [24]

4.3.1 Motivation

Turbine blades are one of the most difficult parts to manufacture on the market today. They are found in the hot section of jet engines and gas turbines, and undergo tremendous loads through their lifetime. In addition to constantly spinning around 5000rpm for hundreds or thousands of hours, they are subject to temperatures in excess of $1500°C$ while spinning. To remedy this problem, manufacturers have spent decades refining the casting process to grow blades that can retain their properties over their service life. In addition to tight material properties and crystal growth control, there are tiny holes going through the blades. These holes are no larger than 1mm in diameter, and go through 6 inches of blade. This kind of intricate detail makes these blades difficult to cast and machine, and lends iself to LPBF, in which tight tolerances can be met.

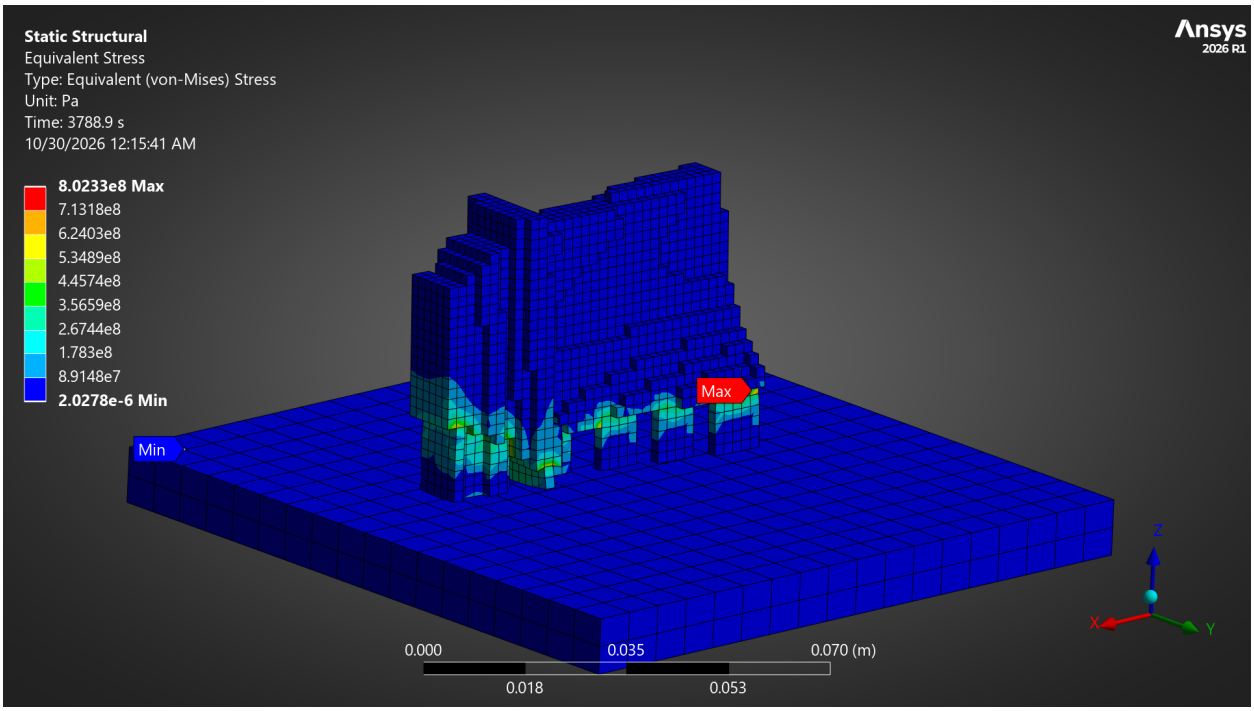


Figure 22: The final stress result of the simulated turbine blade.

This was the final rotation after a full run of stress optimization.

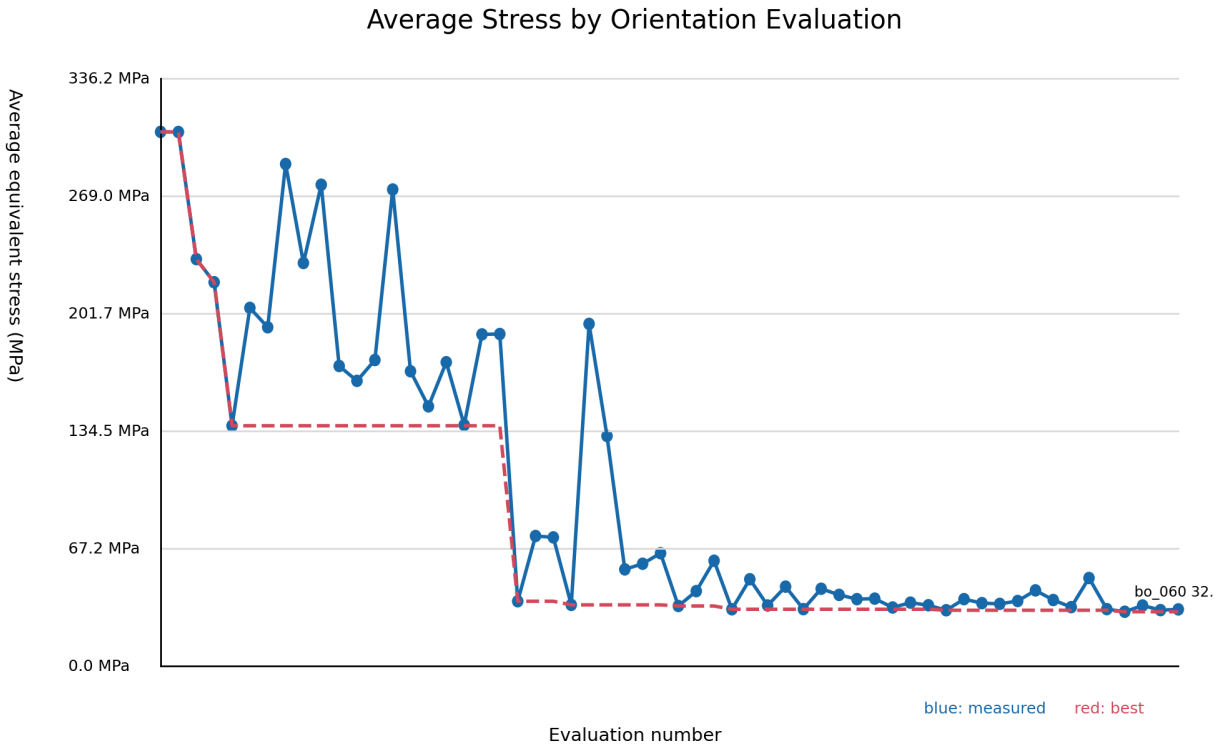


Figure 23: The graph showing the progression of average stress over the run's iterations.

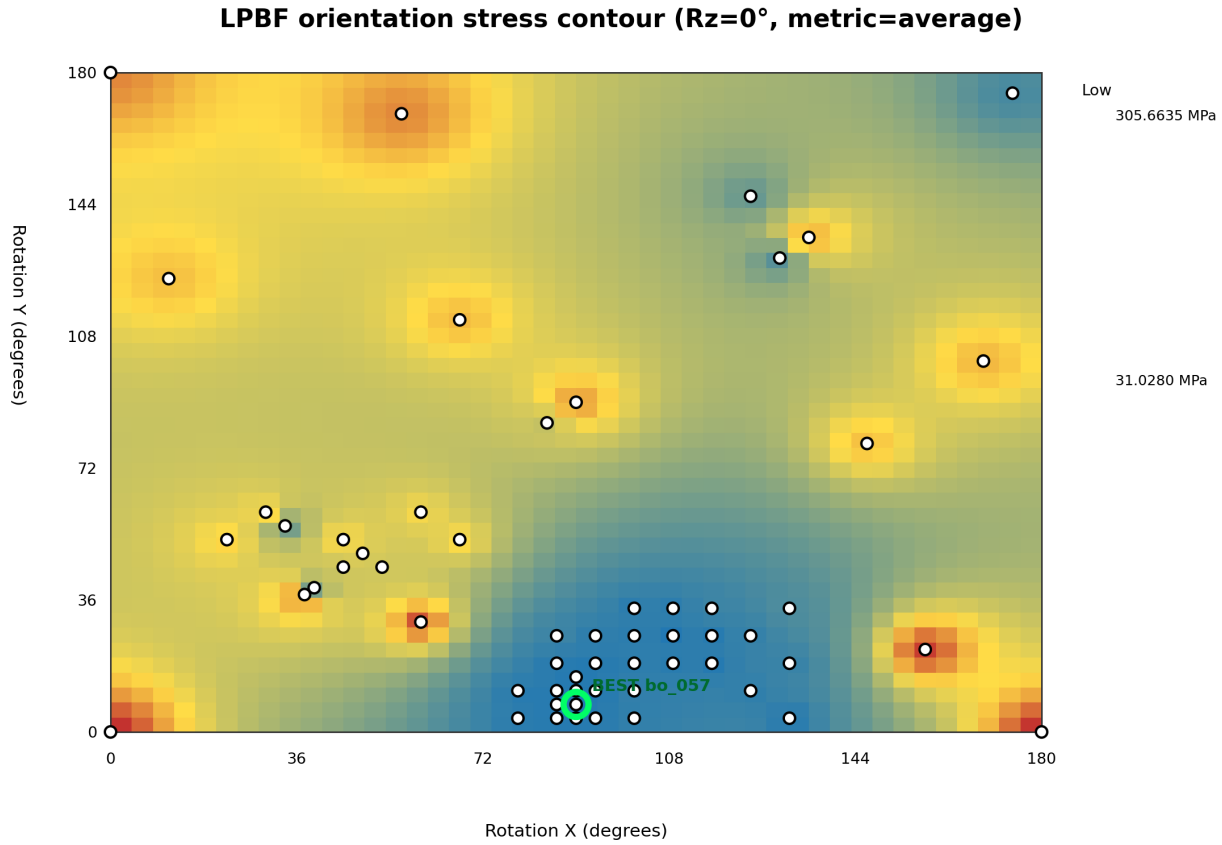


Figure 24: A contour map of the stress vs. rotation angle. Note the data is quite sparse, so graph usefulness may be questionable.

4.3.2 Setup

The geometry was preprocessed according to Section 2, and a mesh of 2mm was chosen, along with 10mm on the base.

4.4 Exhaust Manifold



Figure 25: A 3D printed exhaust manifold [25].

4.4.1 Motivation

Exhaust manifolds are prime examples of a part with difficult geometry for manufacturing along with a high temperature requirement, being located just outside of the combustion chamber of a piston engine. The common method of mass-manufacturing is casting, however, the sand molds need to be multi-piece to create the hollow sections for exhaust gases to flow through. In lieu of mass production, exhaust manifolds are frequently made by cutting and bending pipe segments, then welding them together. This is a process that requires great fabrication skill, and is often an expensive process of any custom engine build. Recently, with this in mind, custom engine fabricators have begun to apply AM to exhaust manifolds, with multiple printable materials being able to handle the high temperatures this part is subject to.

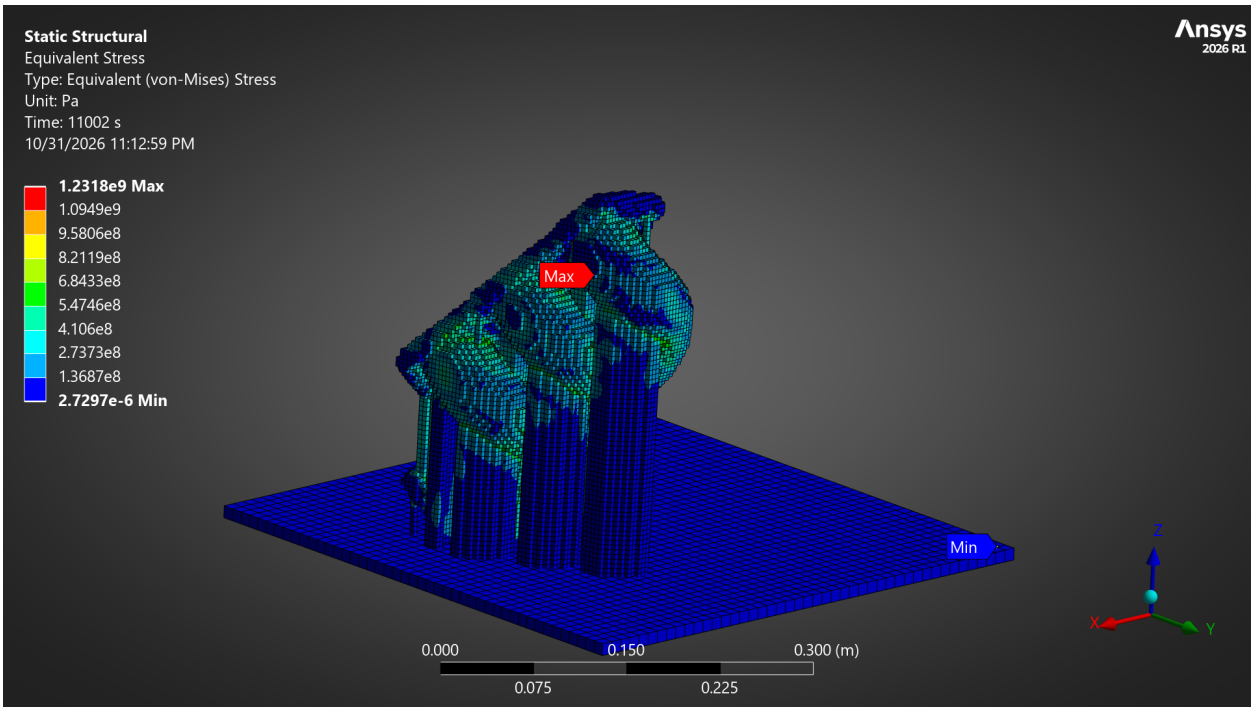


Figure 26: The final stress result of the simulated exhaust manifold.

This was the final rotation after a full run of stress optimization.

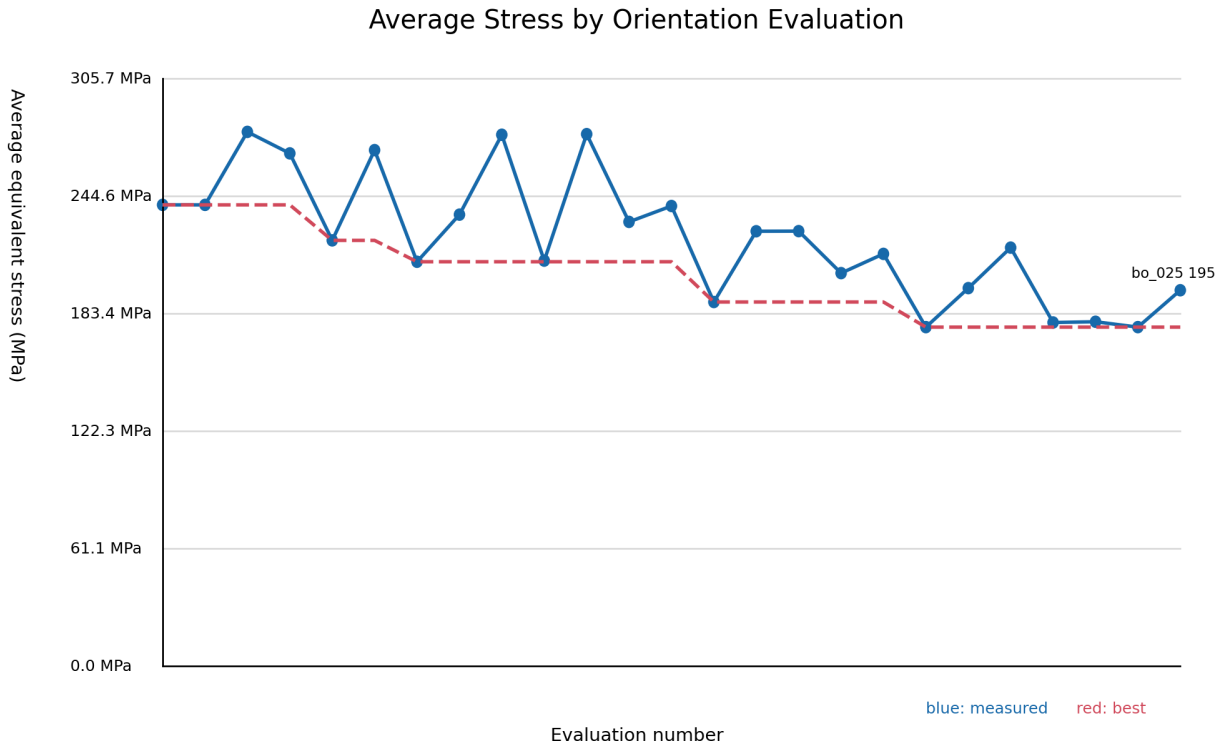


Figure 27: The graph showing the progression of average stress over the run's iterations.

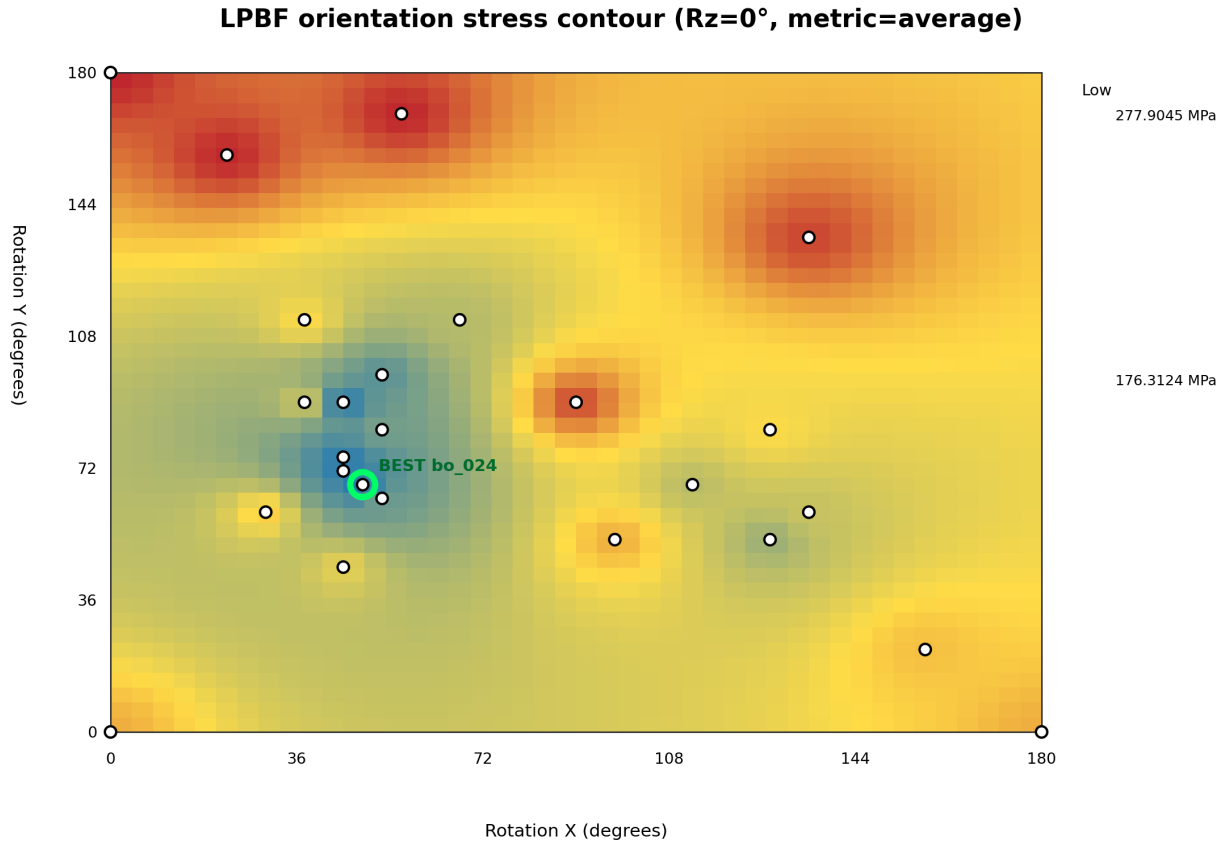


Figure 28: A contour map of the stress vs. rotation angle. Note the data is quite sparse, so graph usefulness may be questionable.

5 Discussion

The results of this project are promising; the code finds an optimal orientation to print parts in given the constraints of compute time. By utilizing methods that are meant for sparse data sets, the compute time is kept relatively low.

However, this simulation only takes into account residual stress. There are no considerations made for dimensional accuracy, print time, support strategy, or material usage. As shown in the exhaust manifold, the direction of print gives greater z height. This typically makes parts take longer to print. Not only this, but printers are typically less accurate in the z direction (check that) so putting more print in that direction could have some tolerance stackup to contend with. With this program, it is completely possible that excessive supports are created. This not only kills print time, but increases material usage greatly. If more supports are used, then more powder was used to print the supports, thus the print cost more. In addition to this, there is no prohibition in the program for printing supports in inaccessible locations. With LPBF, supports need to be removed via wire EDM, which requires an open path to cut. If there is a set of supports inside of a tube for example, wire EDM isn't going to be able to cut them off. In this case, more difficult methods such as machining or grinding could be used, but these are costly and less optimal.

References

[1] M. Tebianian, S. Aghaie, N. R. Jafari, S. E. Hosseini, A. Pereira, F. Fernandes, M. Farbakhti, C. Chen, and Y. Huo, "A review of the metal additive manufacturing processes," *Materials*, vol. 16, no. 24, p. 7514, Dec. 2023.

[2] C.-h. Jung, M. G. Lee, C. Nam, and Y. Jeon, "Numerical analysis of laser pattern effects to residual stress on metal 3d printing," *Applied Sciences*, vol. 12, no. 9, p. 4611, May 2022.

[3] E. De Leon, A. Riensche, B. D. Bevans, C. Billings, Z. Siddique, and Y. Liu, "A review of modeling, simulation, and process qualification of additively manufactured metal components via the laser powder bed fusion method," *Journal of Manufacturing and Materials Processing*, vol. 9, no. 1, p. 22, Jan. 2025.

[4] L. Cai, B. Ghaffari, Y. Huo, M. Li, and S. Y. Liang, "Closed-form full-field modeling of thermal stress in metal powder bed fusion considering part boundary," *Journal of Manufacturing Processes*, vol. 121, pp. 221–232, Jul. 2024.

[5] F. Fojtík, R. Potrok, J. Hajnyš, Q.-P. Ma, L. Kudrna, and J. Měsíček, "Quantification and analysis of residual stresses in braking pedal produced via laser–powder bed fusion additive manufacturing technology," *Materials*, vol. 16, no. 17, p. 5766, Aug. 2023.

[6] S. C. Park, I. Kim, K.-A. Lee, S.-J. Seo, D.-K. Kim, and B. Lee, "Microstructure and residual stress analysis as heat treatment of additive manufactured a356.2 alloy by powder bed fusion," *Materials Characterization*, vol. 207, p. 113538, Jan. 2024.

[7] P. Yang, D. He, X. Guo, S. Lu, S. Chen, F. Shang, D. Oleksandr, and L. Chen, "Effect of printing orientation on the mechanical properties of 3d-printed cu–10sn alloys by laser powder bed fusion technology," *Metals*, vol. 14, no. 6, p. 660, Jun. 2024.

[8] A. Takezawa, H. Guo, R. Kobayashi, Q. Chen, and A. C. To, "Simultaneous optimization of hatching orientations and lattice density distribution for residual warpage reduction in laser powder bed fusion considering layerwise residual stress stacking," *Additive Manufacturing*, vol. 60, p. 103194, Dec. 2022.

[9] H. Alzyod and P. Ficzere, "Material-dependent effect of common printing parameters on residual stress and warpage deformation in 3d printing: A comprehensive finite element analysis study," *Polymers*, vol. 15, no. 13, p. 2893, Jun. 2023.

[10] H. Alzyod, L. Borbas, and P. Ficzere, "Rapid prediction and optimization of the impact of printing parameters on the residual stress of fdm-abs parts using l27 orthogonal array design and fea," *Materials Today: Proceedings*, vol. 93, pp. 583–588, 2023.

[11] S. Zelder, I. Shevchenko, J. Zur Jacobsmühlen, J. Krüger, and E. Uhlmann, "Learning automatic part orientation for powder bed fusion of metal from part data," *Progress in Additive Manufacturing*, vol. 10, no. 3, pp. 1713–1719, Mar. 2025.

[12] B. Weinhold, B. Heck, A. Albright, K. Wang, J.-M. Grote, E. Adeniji, M. Masoomi, and S. Thompson, "Simulation of the laser-powder bed fusion process for determining the effects of part-to-substrate location and orientation on distortion in a connecting rod," in *Solid Freeform Fabrication 2021: Proceedings of the 32nd Annual International Solid Freeform Fabrication Symposium – An Additive Manufacturing Conference*, Austin, TX, 2021, pp. 150–156.

[13] Ľ. Kaščák, J. Varga, J. Bidulská, R. Bidulský, and T. Kvačkaj, "A review of simulation tools utilization for the process of laser powder bed fusion," *Materials*, vol. 18, no. 4, p. 895, Feb. 2025.

[14] P. Rong, Q. Bai, M. Yin, W. Li, Y. Song, and F. Zheng, "Thermal mechanical modeling and residual stress prediction for laser direct energy deposition of 30crni2mova steel considering solid-state phase transformation," *Journal of Materials Research and Technology*, vol. 30, pp. 7727–7740, May 2024.

[15] Dassault Systèmes, "SOLIDWORKS," Computer software, 2024, version 2024.

[16] M. Landry, "Automating solidworks using python," https://mason-landry.hashnode.dev/automating-solidworks-using-python-492b15303db3, Mar. 2022, updated October 24, 2024. Accessed: 2026-05-09.

[17] ANSYS, Inc., "PyMechanical," Python software library, 2026, version 0.12. Available: https://mechanical.docs.pyansys.com/. Accessed: 2026-05-09.

[18] ——, "PyAdditive," Python software library, 2026, version 0.20.0. Available: https://additive.docs.pyansys.com/. Accessed: 2026-05-09.

[19] M. Renardy, L. R. Joslyn, J. A. Millar, and D. E. Kirschner, "To sobol or not to sobol? the effects of sampling schemes in systems biology applications," *Mathematical Biosciences*, vol. 337, p. 108593, Jul. 2021. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8184610/

[20] SciPy Developers. (2026) Sobol — scipy v1.18.0 manual. Documentation for `scipy.stats.qmc.Sobol`. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.qmc.Sobol.html

[21] N. Menon, S. Mondal, and A. Basak, "Multi-fidelity surrogate-based process mapping with uncertainty quantification in laser directed energy deposition," *Materials*, vol. 15, no. 8, p. 2902, 2022. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9025529/

[22] *Ansys LPBF Simulation Guide*, ANSYS, Inc., Canonsburg, PA, Jan. 2025, release 2025 R1. Ansys, Inc., Southpointe, 2600 Ansys Drive, Canonsburg, PA 15317.

[23] AddUp. (2026) Impeller wheel: Optimizing the scan strategy for mass production of an impeller. Automotive metal powder bed fusion case study. [Online]. Available: https://addupsolutions.com/applications/impeller-wheel/

[24] F. Caiazzo, V. Alfieri, G. Corrado, and P. Argenio, "Laser powder-bed fusion of inconel 718 to manufacture turbine blades," *The International Journal of Advanced Manufacturing Technology*, vol. 93, pp. 4023–4031, 2017. [Online]. Available: https://doi.org/10.1007/s00170-017-0839-3

[25] D. Sher. (2020, Dec.) Papadakis racing shows off massive inconel 3d-printed turbo manifold. [Online]. Available: https://www.voxelmatters.com/papadaskis-racing-shows-off-massive-inconel-3d-printed-turbo-manifold/